Science of the Total Environment 782 (2021) 146741

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Occurrence of emerging contaminants in highly anthropogenically influenced river Yamuna in India

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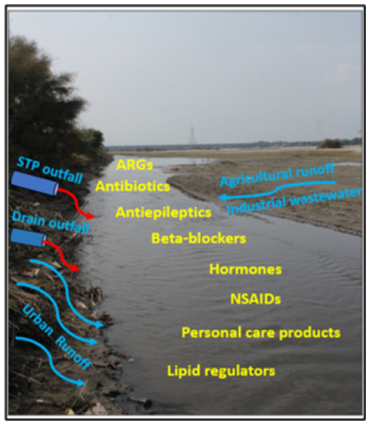
HIGHLIGHTS

- Detection of 16 emerging contaminants (ECs), including pharmaceuticals, personal care products and hormones in river Yamuna.
- Among the 16 target compounds, trimethoprim, an antibiotic, was found in highest concentration followed by caffeine and gemfibrozil, a lipid regulator.
- The concentration of the antibiotic, trimethoprim in river Yamuna is the highest reported in any surface water in the world.
- ECs were also detected in sparsely populated, most upstream sampling location, in the lower Himalayas.
- ECs were detected in the finished drinking water indicating the ineffectiveness of the present water treatment techniques.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 31 December 2020
Received in revised form 20 March 2021
Accepted 21 March 2021
Available online 30 March 2021

Editor: Yolanda Picó

Keywords:
Yamuna
Pharmaceuticals
Personal care products
Endocrine disruptors
Emerging contaminants
Antibiotics

ABSTRACT

River Yamuna is one of the major lifelines of Northern India. The study quantified 16 target compounds including pharmaceuticals, personal care products, and hormones in the Yamuna river. Surface water samples were collected from 13 locations spanning 575 km along the river, and from two of its tributaries, Hindon river and Hindon canal. Spatiotemporal variations in the occurrence of the target compounds at the 13 sites during summer and post-monsoon season were investigated. Caffeine, estrone, gemfibrozil, sulfamethoxazole, testosterone and trimethoprim were found in all the samples, indicating substantial usage and/or persistence in the environment. The mean concentration of the target compounds ranged from 25.5 to 2187.5 ng/L. Higher concentrations were detected during the post monsoon, compared to the summer season. The highest concentration detected was of trimethoprim (8807.6 ng/L) during summer sampling, followed by caffeine (6489.9 ng/L) and gemfibrozil (2991 ng/L), during the post-monsoon sampling. The lowest concentration detected was of estrone (10.7 ng/L), during the summer sampling. The runoff from the catchment areas is one of the contributing factors for the increased concentration of the compounds during post monsoon. During summer, the river bed goes dry, facilitating the adsorption of the compounds onto the river bed sediments. The three sampling locations Okhla barrage (ponding of water from drains traversing Delhi), confluence of Yamuna with Shahadara drain (industrial and poultry

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https://doi.org/10.1016/j.scitotenv.2021.146741
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cluster, and Ghazipur dumping yard) and Agra city (industrial clusters) were the hotspots in terms of total concentration of the target compounds. The study also reported the presence of PPCPs and hormones in the finished drinking water of two DWTPs at Mathura and Agra.

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1. Introduction

The increased consumption of pharmaceuticals and personal care products (PPCPs) has resulted in their ubiquitous presence in the surface water bodies and even in the groundwater aquifers (Ebele et al., 2017, Li et al., 2019, Letsinger et al., 2019, Patel et al., 2019, Andrzejczyk et al., 2020, Yuan et al., 2020). Many of these compounds are known to be endocrine disruptors (Wee et al., 2019). Human metabolism and flushing of unused drugs are some of the pathways through which PPCPs reach the sewerage (Wilkinson et al., 2017). The drugs excreted from the human body after metabolism are mixtures of the parent compounds and their transformation products (Mutiyar et al., 2017). The conventional wastewater treatment facilities are inefficient in the removal of these bio recalcitrant compounds (Hignite and Azarnoff, 1977, Stackelberg et al., 2004, Subedi et al., 2017, Mohapatra et al., 2016). The effluent of the sewage treatment plants with the pharmaceutical residues add to the pharmaceutical load of the surface water sources. The fate of these emerging contaminants (ECs) in the surface water bodies is still unclear. Apart from accumulating in the sediment, aquatic organisms could also be the potential sinks for the PPCPs in the aquatic environment. Wilkinson et al. (2018) detected thirteen PPCPs in the river sediment & biofilms and in the aquatic plants, crustaceans & snails. Quite a few of these emerging contaminants due to their bio recalcitrant nature are remarkably persistent in the environment, thus entering the human food chain (Lagesson et al., 2016; Zhou et al., 2020). The presence of ECs in the drinking water also poses increased health risk to human population. Leung et al. (2013) reported 32 pharmaceuticals in the Chinese tap water samples collected from cities considered contamination hotspots. Tak et al. (2020) detected more than 300 compounds in the drinking water treatment plant (DWTP) at India highlighting the inability of the existing novel water treatment techniques in eliminating ECs from highly polluted sources of drinking water.

The presence of antibiotics in the aquatic environment may lead to the emergence of antimicrobial resistance genes (ARGs) in the bacterial population. Hence, the treatable diseases may become untreatable with the now known antimicrobials due to the presence of resistant genes in the causative bacteria. Rowe et al. (2016) confirmed the contribution of farm and wastewater treatment plant (WWTP) effluent in disseminating ARGs and mobile genetic elements (MREs) in the river catchment of Cambridge, UK. Larsson et al. (2007) reported the highest level of ciprofloxacin in a pharmaceutical WWTP effluent in India. Ahammad et al. (2014) reported the presence of ARGs in the mass bathing sites of the Ganges. The rapid proliferation of antimicrobial resistant infections across the global may lead to high deathrates and increased healthcare costs.

The escalating use of PPCPs presents a grim situation for populous countries such as India. The distribution and fate of ECs in the environmental matrices is still unclear due to lack of comprehensive studies. A few studies have reported the presence of PPCPs in the rivers but are limited to few sampling locations and target compounds. In India the capacity of WWTPs is considerably less than the sewage produced leading to discharge of partially treated or untreated sewage being discharged into the surface water bodies. According to Mohapatra et al. (2016), the attenuation of PPCPs in U.S. WWTPs is 10–30% more compared the Indian WWTPs. In the river, processes such as biological degradation, dilution, sediment adsorption may attenuate the concentration of PPCPs and hormones to a certain extent but the continuous inflow of partially treated or untreated sewage acts an input of the ECs, elevating their concentrations in the surface water which leads to contamination of drinking water and groundwater (Letsinger et al., 2019, Lapworth et al., 2012). Thus, studying the occurrences of PPCPs in the surface waters is the first step towards amelioration of the menace posed by PPCPs.

The main aim of this work is to study the occurrence and distribution of 16 PPCPs and hormones in one of the primary rivers of Northern India, Yamuna, during the summer and post-monsoon seasons. Millions of people depend on Yamuna for their daily water needs. It crosses the highly populated and industrial parts of India before merging with the Ganges. Yamuna flows through Delhi, India's capital which has the highest number of STPs in the country (Mutiyar et al., 2017). This study has tried to highlight the extent of anthropogenic influence on one of the over-exploited rivers of India. It is the first study in India that has covered almost the entirety of a major river (575 km) for the quantification of 16 PPCPs and hormones. Additionally, the results of the summer and post-monsoon seasons were compared to know the extent of attenuation or amplification of concentration due to runoff from the catchment. Since, Yamuna is a source of drinking water, influent and effluent of two DWTPs were also examined for the presence of ECs. The present study also throws light on the inefficiency of the existing water treatment facilities to completely remove the compounds of emerging concern from the potable water.

2. Materials and method

2.1. Chemicals and materials

16 compounds were chosen for this study including pharmaceuticals, personal care products and endocrine disruptors (Table 1). The list of the compounds with their CAS numbers and supplier of the standards are given in supplementary information (S-1). All the standards used in this study are of highest purity grade. All the solvents used for SPE and HPLC-MS analysis were of HPLC grade except for water. Solvents used for UPLC-QToF analysis were of LCMS grade. Solvents used for the preparation of stock solution were purchased from J.T. Baker. LCMS grade water was used throughout the study and was obtained from Evoqua water systems. Acetic acid, ammonium acetate, formic acid and hexane were of HPLC grade with >98% purity. 100 ppm stock

Table 1
List of the studied compounds their respective classes:

Compounds	Class
Caffeine	Stimulant
Trimethoprim	
Ciprofloxacin	Antibiotic
Sulfamethoxazole	
Enrofloxacin	
Testosterone	
Progesterone	Hormone
Estrone	
Diclofenac	
Naproxen	NSAID
Ketoprofen	
Carbamazepine	Anticonvulsant
Atenolol	Beta blocker
Triclosan	Antibacterial and antifungal
Gemfibrozil	Lipid Regulator
2-Hydroxybenzothiazole	Vulcanization accelerator

solutions of individual standards were prepared in methanol except for caffeine, atenolol and trimethoprim which were prepared in LCMS grade water and were stored in the dark at −20 °C. Ciprofloxacin was prepared in 50 mM aqueous orthophosphoric acid solution. Working solutions were prepared from these stock standard solutions with appropriate dilutions. SPE of surface water and drinking water were done using Waters 12 position vacuum manifold and Oasis HLB cartridges (6 cc, 200 mg). Rapid mini nitrogen evaporator purchased from Crescent Scientific was used for drying the samples. Surface water analysis was carried out with Waters Alliance Liquid Chromatography System coupled with Waters SQ mass detector. Nitrogen used for drying and nebulizing was supplied by nitrogen generator purchased from Peak Scientific. Drinking water screening was carried out with UPLC (Agilent 1290 infinity II) coupled to QToF (Agilent 6550 QToF). National Institute of Standards and Technology (NIST) library was used as the database for screening in UPLC-QToF.

2.2. Study area

The river Yamuna rises from the Yamunotri glacier in the Himalayas and the river basin spreads across an area of 366,220 sq.km. Surface water samples were collected along the river Yamuna and its tributaries, with sampling points concentrated near clusters human settlements. 13 sites were selected at different locations covering a distance of 575 km from Dakpatthar in the lower Himalayas, the most upstream sampling point, to Etawah, the most downstream sampling point. The first sampling point Dakpatthar (Y-1) is situated close to the origin of Yamuna at the lower Himalayas with a sparse population. After Y-1, the next three sites were selected in and around the capital, Delhi, including Nizamuddin bridge (Y-4), Okhla barrage (Y-5) and downstream of Yamuna-Shahadara drain confluence (Y-6). The chosen sites in Delhi are situated at the downstream of the confluence points of major and minor drains with Yamuna. At least 12 drains including Najafgarh drain with a BOD load of 132.55 t/day have their outfalls at the upstream of site Y-4 (*Annual Average flow and BOD load of drains*, 2017). At Y-5, domestic and industrial waste from Gaziabad city flows into the Yamuna (Mutiyar et al., 2017). Y-6 is approximately 1 km downstream of Shahadara drain and Yamuna confluence. Sharadara drain, which flows through the industrial townships of Delhi, brings BOD load of 41.26 t/day into the Yamuna (*Annual Average flow and BOD load of drains*, 2017). Water samples were also collected from Hindon river (Y-2) and Hindon canal (Y-3) which act as the tributaries of Yamuna. Hindon river is a rainfed river which joins Yamuna in the outskirts of southern Delhi. Hindon river branches into Hindon canal and joins Yamuna in Delhi. Both Hindon river and Hindon canal flow through highly industrialized and populated areas before joining Yamuna. In order to understand the effect of Mathura, a town with approximate 442 thousand population, of great religious importance, samples were collected upstream (Y-7) and downstream of Mathura (Y-8). Next three sites were chosen from Agra, a town with approximately 1.59 million population and a tourist hotspot. Samples upstream of Agra (Y-9), Agra city near Taj Mahal (Y-10) and downstream of Agra (Y-11) were collected. The last two sites were chosen upstream (Y-12) and downstream of Etawah (Y-13). The site Y-13 is close to the outfall of the Etawah STP. Etawah is town of 257 thousand population with a basic WWTP based on waste stabilization pond. All the 13 surface water sampling locations are represented in Fig. 1.

Fig. 1. 13 selected surface water sampling locations including 11 sites along Yamuna river and 2 sites along its tributaries namely Hindon river and Hindon canal resp. Location Y-1: Dakpatthar in inset I, location Y-2: Hindon river, Y-3: Hindon canal, Y-4: Nizamuddin Bridge, Y-5: Okhla Barrage and Y-6: D/S Shahadara drain & Yamuna Confluence in inset II, location Y-7: U/S Mathura and Y-8: D/S Mathura in inset III, location Y-9: U/S Agra city, Y-10: Agra city near Taj Mahal and Y-11: D/S Agra city in inset IV, location Y-12: U/S Etawah and Y-13: D/S Etawah in inset V.

For drinking water sampling, two DWTPS at Mathura and Agra were selected. Both the plants have inlets in Yamuna. Mathura DWTP is based on conventional water treatment technique while Agra DWTP is based on moving bed biofilm reactor (MBBR) method. The schematics of both the DWTPs are given in the supplementary information (S-2).

2.3. Sampling

Grab samples were collected from a depth of 0.5 m from the surface by a stainless-steel manual weighted surface water sampler. The surface water samplings were carried out during summer and post-monsoon season. The summer sampling was conducted in the month of May while post-monsoon sampling was conducted during September few days after a major flooding. The average monthly discharges of water in Yamuna from two barrages situated at upstream of Delhi are given in the supplementary information (S-3). The general water quality parameters were given in supplementary information (S-4). Grab samples were collected from inlet and outlet of both Mathura and Agra DWTP. The samples were collected in 2 L amber glass bottles and maintained at a temperature of 4 °C throughout the transportation. The samples were first vacuum filtered through 0.45 μm and 0.2 μm filter papers and then extracted by solid phase extraction. The method for solid phase extraction is given in the supplementary information (S-5). The surface water samples were analyzed by HPLC-MS while the inlet and outlet samples from the two DWTPs were analyzed by UPLC-QToF. The methods used for HPLC-MS and UPLC-QToF are given in supplementary information (S-6).

3. Results and discussion

3.1. Optimization of solid phase extraction

For the optimization of SPE, water from Dakpatthar (close to the origin of Yamuna) was spiked with stock solution containing mix of the 16 target analytes, to get a final concentration of 5 μg/L each. The % absolute recovery (R) of each trial was calculated from the following equation.

$$R(\%) = \frac{A_{pre-SPE\ spiked} - A_{non-spiked}}{A_{non-SPE\ std}} \quad (1)$$

where $A_{pre\text{-}SPE\ spiked}$ is the peak area of the analyte in matrix spiked before SPE, $A_{non\text{-}spiked}$ is the peak area of the analyte in the non-spiked matrix and $A_{non\text{-}SPE\ std}$ is the peak area of the analyte in the standard solution without SPE (Vieno et al., 2006). Based on published literature, hydrophilic-lipophilic balanced copolymer cartridges (Oasis HLB, 200 mg, 6 cc) were used (De Oliveira et al., 2019; Zhang and Zhou, 2007; Vieno et al., 2006). For SPE recovery trials, 500 mL of sample volume was used. The first trial of solid phase extraction was referred from EPA (*Method 1694: Pharmaceuticals and Personal Care Products in Water, Soil, Sediment, and Biosolids by HPLC/MS/MS*, 2007). The samples were acidified and spiked before SPE with 16 analytes. Apart from poor recoveries for all the target compounds except estrone and enrofloxacin, the analysis also showed deviation in RTs which could be because of the use of methanol as reconstitution solvent. The RT for each compound was fixed from multiple injections of the standard prepared by diluting the stock solution of the compound in aqueous media. Since, methanol is a non-polar matrix relative to water, there was mismatch of retention time in the first trial (R_1).

The deviation in RT was addressed in the second trial (R_2). However, in order to save the time of analysis, out of 16 only 6 compounds with varying degree of polarities were chosen. In this trial, the reconstitution solvent was changed to 80% aqueous instead of 100% methanol. The analysis showed no deviation in RTs. However, the % recoveries of the compounds were still poor, except for carbamazepine. The third trial (R_3) was performed aiming to improve the recovery as well as to simplify the sample preparation by keeping the sample close to neutral pH. The change in the sample pH had positive effects on the recoveries of few of the compounds which suggested that the % recoveries were closely associated with the sample pH. In the 4th trial (R_4), both acidic and basic samples sets were prepared for all the 16 compounds. Each set was spiked with all the compounds before SPE. The target compounds were eluted in two steps. One was done by 8 mL acidified (0.1% acetic acid) methanol followed by a 6 mL of mixture of isopropyl alcohol and acetonitrile (50:50). In the next step, elution was completed with 3 mL of hexane in a separate test tube. Thus, solvents of different polarities were used as elution solvents. The percentage recoveries improved in both acidic and basic samples except for triclosan. In the fifth trial (R_5), the wash step was discontinued as it can cause loss of hydrophilic compounds such as caffeine. The percentage recoveries of all the five trials are presented in Table 2.

Two field blanks containing LCMS grade water each for the two sampling sessions were analyzed for the target analytes. Similarly, four laboratory blanks containing the mobile phases were investigated for the analytes during every set of HPLC-MS and UPLC-QToF run. The analysis of both the field and laboratory blanks showed that the laboratory wares and the mobile phases were free from contamination with the target analytes. Nine points linear calibration curves for each analyte were constructed by using standard solutions with concentrations

Table 2
Absolute recovery (%) of the target analytes during the trials.

Compounds	R_1	R_2	R_3	R_4		R_5	
				Acidic condition	Basic condition	Acidic condition	Basic condition
Estrone	107	58	67	102		102	
Ciprofloxacin	2	nd	4	139		120	
2-Hydroxybenzothiazole	2	nd	nd	64		89	
Ketoprofen	44	nd	nd		62		74
Sulfamethoxazole	2	nd	25	82		91	
Atenolol	Deviation of RT by 2 min	2	23		56		75
Testosterone	21	nd	16		80		80
Carbamazepine	9	91	18	52		54	
Progesterone	11	nd	nd	104		100	
Diclofenac	2	nd	nd		26		78
Triclosan	24	nd	49	nd	nd	64	
Caffeine	18	28	32		68		95
Naproxen	23	11	20		70		77
Gemfibrozil	16	nd	nd		89		91
Enrofloxacin	79	nd	nd		110		94
Trimethoprim	7	1	11		90		89

R denotes percent recovery in respective trials.

ranging between 1 and 650 μg/L. Method detection limit (MDL) and Method quantification limit (MQL) for each target compound was calculated according to EPA. MDL was calculated by analyzing seven spiked samples and method blanks. MQL was found by multiplying MDL with a factor of 3.18. Both MDL and MQL are presented in supplementary information (S-6). To check the instrument response variability, multiple injections were given for each and every sample analyzed by HPLC-MS.

3.2. Detection of the target analytes in surface water

3.2.1. Frequency of detection

The occurrence of the PPCPs and hormones in the Yamuna during the summer and post-monsoon season are summarized in Table 3. Out of the 16 target compounds, all of them have frequencies of detection greater than 20%. 13 of the 16 compounds have frequencies higher than 80% with caffeine, estrone, gemfibrozil, sulfamethoxazole, testosterone and trimethoprim being detected in all the samples. The highest concentration of PPCP detected was that of trimethoprim at 8807.6 ng/L at Okhla barrage (Y-5) within Delhi, during post-monsoon season. The presence of trimethoprim is consistent with the findings of Murata et al. (2011) who had reported its occurrence in Japanese rivers. Trimethoprim along with sulfonamides are one of the most preferred antibiotics for the treatment of common infections (*WHO Report on Surveillance of Antibiotic Consumption*, 2016–18). This is also evident from the presence of sulfamethoxazole in all the surface water samples. The heavy usage of trimethoprim is reflected in its high concentration reported by Balakrishna et al. (2017) and Joshua et al. (2020) in the surface waterbodies of southern India. The second highest concentration was of caffeine (6489.9 ng/L), at Y-7 (downstream of Shahadara drain-Yamuna confluence) during post-monsoon season. Mutiyar et al. (2017) reported 2082 ng/L of caffeine in the Yamuna samples taken from upstream of Delhi. However, most of the samples in the present study are taken from the urban areas and such high concentration of caffeine indicates mixing of wastewater effluents. The average caffeine intake in Asia is about 165.87 mg/person/day, a lot less then U.S. and Europe (Korekar et al., 2020). In spite of its biodegradability, extensive consumption of caffeine has led to consistent detection in the rivers around the globe and therefore caffeine acts as an indicator of anthropogenic contamination (Brumovský et al., 2017; Korekar et al., 2020). Caffeine concentrations ranging from 131 ng/L to 5540 ng/L was reported by Škrbić et al. (2018) in Serbian surface water. High levels of caffeine equal to 7630 ng/L was also confirmed in a study conducted on Slovenian wastewater (Česen et al., 2018). Other than trimethoprim and sulfamethoxazole, the antibiotics ciprofloxacin and enrofloxacin, were also found in high frequencies (<90%), indicating widespread usage of different classes of antibiotics. Kuang et al. (2020) reported 48 antibiotics in the urban river of China. The presence of hormones at almost all the sampling stations points to the addition of human and animal excreta. The excretion of hormones from human and animal sources had also been discussed by Barel-Cohen et al. (2006) while monitoring lower Jordan river for natural and synthetic waste. The study associated agricultural and fish pond effluent with the presence of estrone and testosterone in some sections of the river. 2-hydroxybenzothiazole has been detected in almost all the samples except Y-5 during summer sampling (96%). Benzothiazoles have industrial as well as everyday application and its consistent presence in the water samples indicates leaching from the end products (Mazioti et al., 2015).

3.2.2. Spatiotemporal variation of PPCPS and hormones

The occurrence of the 16 target compounds along the sampling locations during summer and post-monsoon seasons is presented in Fig. 2.

The mean concentration of the detected target compounds ranged between 25.5 and 2187.55 ng/L. The concentrations of caffeine, naproxen, enrofloxacin, gemfibrozil, carbamazepine, atenolol, sulfamethoxazole, 2-hydroxybenzothiazole, diclofenac, ciprofloxacin, progesterone and triclosan showed clear distinctions between the two seasons. Unlike the dry season, Yamuna receives substantial volume of water during the monsoon. As a result, the barrages that held the water during summer acting as oxidation ponds release the contaminant laden water during monsoon, deteriorating the downstream river water quality. A few days before the post-monsoon sampling, the catchment area witnessed considerable rainfall leading to runoff into Yamuna. The runoff from the catchment areas aggravates the already deteriorating quality of the Yamuna water. The effect of the runoff is evident from the elevated post-monsoon detection patterns of atenolol, caffeine, carbamazepine, ciprofloxacin, diclofenac, enrofloxacin, gemfibrozil, ketoprofen, naproxen, progesterone, sulfamethoxazole and 2-hydroxybenzothiazole at the sites situated in Delhi and Agra city (Y-4, Y-5, Y-6 & Y-10). All these compounds were detected in much higher concentration in the post-monsoon samples collected from Nizamuddin bridge (Y-4), Okhla barrage (Y-5), downstream of the confluence of Yamuna & Shahadara Drain (Y-6) and Agra city (Y-10). Boyd et al. (2004) identified the similar phenomena due to 'First flush' in the investigation of PPCPs and EDCs in the New Orleans stormwater. Such high concentrations of ECs in the Yamuna can be detrimental to the aquatic lifeforms (Gomes et al., 2020). Bioaccumulation and biomagnification may play significant role in transferring the effects of long-term exposure to ECs from the aquatic organisms to the terrestrial organisms via food chain (Zenker et al., 2014).

Apart from runoff, Yamuna dries up drastically during the summer season, which may lead to accumulation of the compounds in the

Table 3
Detection of PPCPs in the Yamuna river water.

Compounds	Frequency of detection (%)	Maximum conc. (ng/L)	Minimum conc. (ng/L)	Mean conc. (ng/L)
Caffeine	100	6489.9 ± 0.8	74.8 ± 1.2	707.9
Naproxen	88	1576.3 ± 1.9	nd	340.8
Enrofloxacin	92	59.8 ± 2.8	nd	40.2
Gemfibrozil	100	2990.4 ± 49.9	48.2 ± 3.3	792.4
Testosterone	100	284.5 ± 1.5	52.6 ± 1.2	113.6
Carbamazepine	88	484.1 ± 1.4	nd	119.2
Sulfamethoxazole	100	1312.3 ± 0.1	48.5 ± 1.2	183.5
Atenolol	38	205.6 ± 1.8	nd	44.4
Ketoprofen	77	1381.2 ± 1.8	nd	133.7
Estrone	100	1781.8 ± 0.7	10.7 ± 0.4	476.6
2-Hydroxybenzothiazole	96	591.9 ± 1.4	nd	216.4
Progesterone	85	291.3 ± 1.4	nd	95.1
Triclosan	38	269.8 ± 1.0	nd	25.5
Diclofenac	85	278.9 ± 0.7	nd	96.1
Trimethoprim	100	8807.6 ± 2.7	64.5 ± 1.3	2187.5
Ciprofloxacin	96	263.5 ± 1.7	nd	115.1

nd denotes not detected; mean concentration of a compound is the sum of the concentrations at the sampling locations divided by number of locations.

river bed sediments. Rainfall caused scouring increased the turbulence in the river possibly leading to desorption of the compounds into the aqueous phase. The high TSS concentration of the post-monsoon samples (supplementary information S-4) backs this approach. The present pattern is comparable to the results from a study on the Kyungahn stream in South Korea. The study found increased detection rate and concentration of tetracyclines and sulfonamides in the stream water during the rainy season. The dry Kyungahn stream during summer accumulates the compounds in the sediments while releasing them once the stream flow increases during monsoon (Kim et al., 2016).

Triclosan was not detected in any of the post-monsoon samples. Due to its extensive application as an antimicrobial agent triclosan is ubiquitous in the aquatic environment (Singer et al., 2002). During the summer months, most of the water in the Yamuna comprises of treated, partially treated or raw wastewater and hence triclosan was detected in most of the summer samples while post monsoon, due to dilution the concentration went down to less than the detection limit. However, triclosan at sub-lethal concentrations in the rivers may trigger the spread of antibiotic resistance (Lu et al., 2018). The concentration of trimethoprim was way higher during the summer sampling. The spiked levels of trimethoprim at the sites Y-4 and Y-5 during summer season may be linked to extreme ponding of WWTP effluents on the river bed.

Unlike triclosan, atenolol was absent in the summer season samples which may be attributed to the photolytic degradation of the compound rendering the concentration below the minimum detection limit. Consideration of the consumption pattern of the compounds during seasons has to be taken into account. Research on the photochemical degradation of contaminants in natural wastewater also confirms the attenuation of atenolol under solar radiation (Dong et al., 2015). Daneshvar et al. (2010) reported a similar attenuation of atenolol in the Swedish river Fyris during summer sampling. Andrisano et al. (1999) confirmed the degradation of atenolol under UV radiation at decreasing pH. The photodegradation of atenolol resulted in the formation of a hydrophilic and a hydrophobic product at varying pH of the sample. According to Dong et al. (2015) sensitized photolysis at the surface of the wastewater causes the direct photolytic removal of atenolol. Based upon the structure of atenolol, reaction with reactive intermediates such as singlet oxygen produced from effluent organic matter could be a possible degradation pathway.

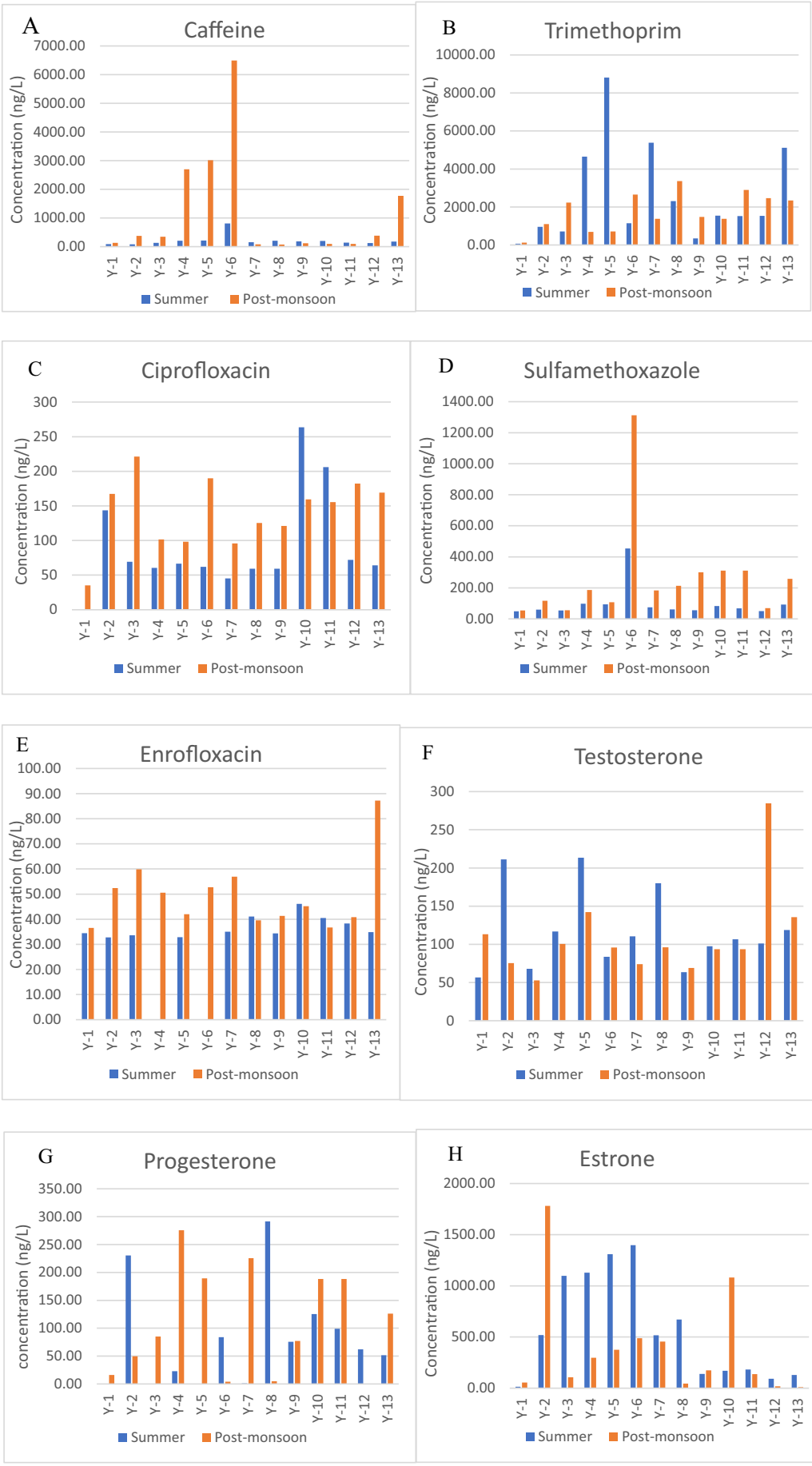
The detection of estrone and testosterone failed to follow any particular pattern. Hence, the effects of runoff on the distribution of the two hormones were unclear. However, when compared with progesterone and testosterone, estrone was detected in greater concentrations which may be linked to its bio recalcitrant nature. Estrone is also the transformation product of natural estrogens, 17ß-estradiol and 17α-ethnyl estradiol (Ternes et al., 1999, Johnson and Williams, 2004).

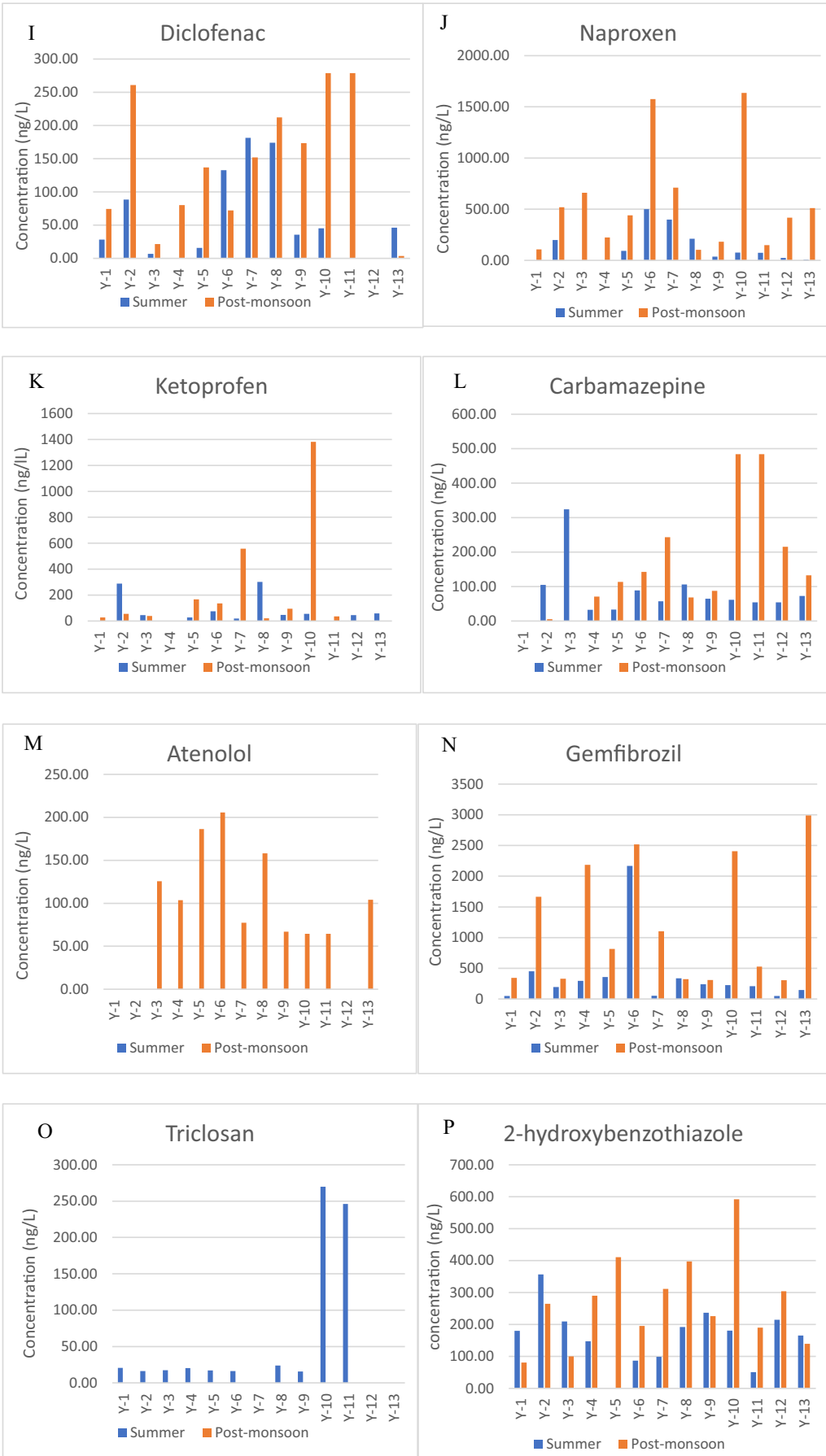
Cumulative concentrations of the 16 target compounds along the 13 sampling locations during summer and post-monsoon is presented in Fig. 3.

The total concentration of the ECs at site Y-1 was much lower compared to the other sampling locations, which may be attributed to the sparsely populated Lower Himalayas near the origin of Yamuna. However, the detection of widely used PPCPs at site Y-1 raises the concern of anthropogenic contamination of Himalayan surface water bodies. The present detection results revealed Okhla barrage, Shahadara drain and Agra city as the hotspots of the target contaminants in the Yamuna. The obstruction of the natural river flow at the Okhla barrage leads to the mixing and retention of the raw sewerage, WWTP effluents and sludge at the upstream of the barrage resulting in higher cumulative concentration of the target compounds. During post-monsoon, the high river flow dilutes most of the compounds at the Okhla Barrage, thereby lowering the cumulative concentrations which is evident from the decreasing trend of estrone, testosterone, triclosan and trimethoprim at Y-5. Sampling location Y-6 became the site with highest cumulative concentration of the target compounds during the post-monsoon. The samples collected from this site contained highest concentrations of caffeine and sulfamethoxazole, second highest concentration of naproxen and gemfibrozil among all the analyzed samples. Sulfamethoxazole is one of the widely used antibiotics against infectious diseases and the concentration quantified in this paper is the highest in India and, to the knowledge of the author, probably in the world. Concentration of 500 μg/L in river water has been seen to promote antibiotic resistance in certain bacterial communities (Patrolecco et al., 2018). Diwan et al. (2009) detected sulfamethoxazole in hospital effluent, however the detected concentration was quite low as compared to the present study. Similar to sulfamethoxazole, naproxen was also detected in high concentrations. The results also showed a decreasing trend in the cumulative concentration of the compounds beyond the Delhi stretch (Y-2 to Y-6) until it spiked at Agra city. Due to the presence of a barrage at the upstream of Agra (at Mathura), the river takes the form of a small stream during the dry season carrying diluted sewerage and only receives water during monsoon. Increased aromaticity of Yamuna water at Mathura following the high values of UV254 has been discussed by Tak and Vellanki (2019). The summer samples collected from Agra city contained elevated concentration of trimethoprim and ciprofloxacin, while the post-monsoon samples had high concentrations of 2-hydroxybenzothiazole, carbamazepine, diclofenac, estrone, gemfibrozil, ketoprofen, naproxen and trimethoprim. The concentrations of 2-hydroxybenzothiazole, diclofenac and ketoprofen were the highest at Agra city among all the analyzed samples indicating the effects of local manufacturing and consumption. Many published papers have found that well-functioning STPs are able to partially eliminate NSAIDs from sewage (Lin et al., 2009; Matamoros et al., 2009; Kosjek et al., 2007). The high concentrations of non-steroidal anti-inflammatory drugs (NSAIDs) in the Yamuna water showed the inefficiency of the existing wastewater treatments in attenuating the NSAIDs from the domestic and industrial sewage. Stepnowski et al. (2020) investigated the presence of NSAIDs in the Vistula river water after the failure of Warsaw WWTP. Elevated concentration of anti-inflammatory compounds such diclofenac, naproxen, etc. were found in the Vistula river water. Singh and Suthar (2021) confirmed the presence of NSAIDs such as diclofenac and ketoprofen in the river Ganges but in much lower concentration. After Agra city, the cumulative compound concentration again showed a slight increase downstream of Etawah (Y-13) indicating the ineffectiveness of the waste stabilization pond in removing ECs. The samples collected at this site during the post-monsoon sampling contained highest concentration of enrofloxacin and gemfibrozil. The concentration of enrofloxacin found in Y-13 is 87.2 ng/L higher than the mean concentration of 40.2 ng/L across all the sites during both seasons. Since enrofloxacin is primarily used as veterinary antibiotics, its concentration in Yamuna is much less than the antibiotics used in humans. With low water solubility enrofloxacin has the tendency to accumulate in the soil and riverbed sediments (Slana et al., 2017). Gemfibrozil is a prescription drug used as lipid regulator. It was detected at 2990.4 ng/L during the post-monsoon sampling. Such elevated levels of gemfibrozil were only reported in wastewater treatment effluents and never in surface water. Bulloch et al. (2015) detected gemfibrozil equal to 2970 ng/L in one of the wastewater treatment effluent in Southern California. Exposure to high concentrations of gemfibrozil may lead to endocrine disruption in fishes (Mimeault et al., 2005).

The study also explored the occurrences of the ECs in the two tributaries to better understand their effect on the Yamuna water quality. Since Hindon river covers a major area, the samples collected from site Y-2 presented high cumulative concentration of the target compounds

Fig. 2. Spatiotemporal variations in the occurrence of the 16 targeted PPCPs and hormones in the Yamuna river water. (Y-1: Dakpatthar, Y-2: Hindon River, Y-3: Hindon Canal, Y-4: Nizamuddin Bridge, Y-5: Okhla Barrage, Y-6: D/S Shahadara drain & Yamuna Confluence, Y-7: U/S Mathura, Y-8: D/S Mathura, Y-9: U/S Agra city, Y-10: Agra city near Taj Mahal, Y-11: D/S Agra city, Y-12: U/S Etawah, Y-13: D/S Etawah).


A
Caffeine
Concentration (ng/L)
7000.00
6000.00
5000.00
4000.00
3000.00
2000.00
1000.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer Post-monsoon
B
Trimethoprim
Concentration (ng/L)
10000.00
8000.00
6000.00
4000.00
2000.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer Post-monsoon
C
Ciprofloxacin
Concentration (ng/L)
300
250
200
150
100
50
0
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer Post-monsoon
D
Sulfamethoxazole
Concentration (ng/L)
1400.00
1200.00
1000.00
800.00
600.00
400.00
200.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer Post-monsoon
E
Enrofloxacin
Concentration (ng/L)
100.00
90.00
80.00
70.00
60.00
50.00
40.00
30.00
20.00
10.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer Post-monsoon
F
Testosterone
Concentration (ng/L)
300
250
200
150
100
50
0
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer Post-monsoon
G
Progesterone
concentration (ng/L)
350.00
300.00
250.00
200.00
150.00
100.00
50.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer Post-monsoon
H
Estrone
Concentration (ng/L)
2000.00
1500.00
1000.00
500.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer Post-monsoon


I
Diclofenac
Concentration (ng/L)
300.00
250.00
200.00
150.00
100.00
50.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer
Post-monsoon
J
Naproxen
Concentration (ng/L)
2000.00
1500.00
1000.00
500.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer
Post-monsoon
K
Ketoprofen
Concentration (ng/lL)
1600
1400
1200
1000
800
600
400
200
0
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer
Post-monsoon
L
Carbamazepine
Concentration (ng/L)
600.00
500.00
400.00
300.00
200.00
100.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer
Post-monsoon
M
Atenolol
Concentration (ng/L)
250.00
200.00
150.00
100.00
50.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer
Post-monsoon
N
Gemfibrozil
Concentration (ng/L)
3500
3000
2500
2000
1500
1000
500
0
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer
Post-monsoon
O
Triclosan
Concentration (ng/L)
300.00
250.00
200.00
150.00
100.00
50.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer
Post-monsoon
P
2-hydroxybenzothiazole
concentration (ng/L)
700.00
600.00
500.00
400.00
300.00
200.00
100.00
0.00
Y-1 Y-2 Y-3 Y-4 Y-5 Y-6 Y-7 Y-8 Y-9 Y-10 Y-11 Y-12 Y-13
Summer
Post-monsoon

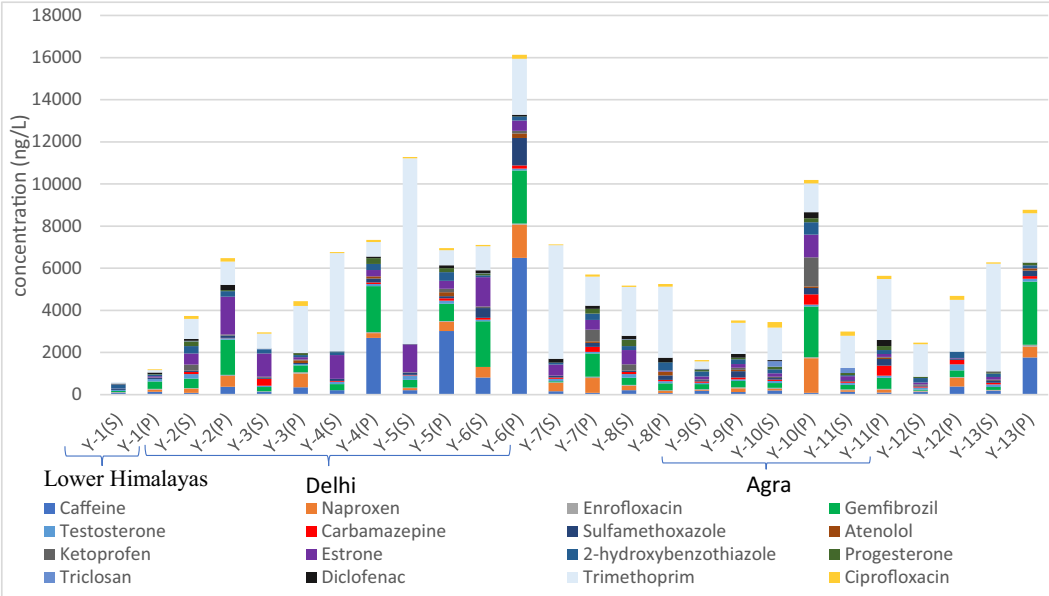


Fig. 3. Spatiotemporal variations in the cumulative concentration of all the 16 targeted PPCPs and EDCs in the Yamuna river water (S stands for summer concentration and P stands for post-monsoon concentration). (Y-1: Dakpatthar, Y-2: Hindon River, Y-3: Hindon Canal, Y-4: Nizamuddin Bridge, Y-5: Okhla Barrage, Y-6: D/S Shahadara drain & Yamuna Confluence, Y-7: U/S Mathura, Y-8: D/S Mathura, Y-9: U/S Agra city, Y-10: Agra city near Taj Mahal, Y-11: D/S Agra city, Y-12: U/S Etawah, Y-13: D/S Etawah.)

than Y-3 (Hindon canal). Hindon river contained the highest concentration of estrone among all the locations which may be linked to the nearby agricultural runoff. According to studies, domestic and industrial wastewater, animal excreta, agricultural runoff, landfills can release estrone in the aquatic environment (Coors et al., 2003, Wise et al., 2011, Shappell et al., 2021). Samples from site Y-3 also contained high concentration of trimethoprim which may be related to their non-point release from the nearby shanties, dairies, fish and livestock markets. The Yamuna tributaries, especially Hindon river, dilutes the contaminant load of Yamuna originating from Delhi which is evident from the decreased total concentration of the ECs at the site Y-7.

3.3. Detection of the target analytes in DWTPs

The distribution of ECs in the inlet and outlet of the two DWTPs are presented in Fig. 4.

The samples collected from the inlet of Mathura DWTP contained 190 ECs out of which 103 were pharmaceuticals, 6 were personal care products and 12 were steroids. While the number of pharmaceuticals were reduced to 86 in the outlet of the DWTP, personal care products and steroids remained unchanged. Tak et al. (2020) reported high concentration of ammonia in the Mathura and Agra stretch of Yamuna due to the effluents released from plastic and petrochemical industries. The heavy anthropogenic contamination of Yamuna at Agra is reflected in the detection of 217 ECs in the samples collected from the Agra DWTP. Out of the 217 compounds, broadly, 144 were pharmaceuticals, 6 were personal care products and 7 were steroids. The MBBR installed at the Agra DWTP was ineffective in removing the ECs from the influent or decreasing the ECs below the detection limit of the UPLC-QToF. The finished drinking water at the Agra DWTP contained 116 pharmaceuticals, 4 personal care products and 6 steroids. Though the decrease in the peak area indicates decrease in concentration of the ECs in the effluents. Both the technologies were unable to effectively attenuate the number of ECs in the finished drinking water.

4. Conclusion

The present study investigated the presence of 16 PPCPs and hormones along a major length of the river Yamuna and two of its tributaries. Comparing the detection pattern of the compounds in Dakpatthar, in the lower Himalayas, with that of the other downstream locations highlighted the extent of anthropogenic influence on the Yamuna water quality. The presence of antibiotics revealed the exploitation of over-the-counter antibiotics and the risk of antimicrobial resistance. The presence of the antibiotics and NSAIDs in the Yamuna is also alarming due to the fact that most of the urban settlements along the river basin are dependent upon the river for their drinking water needs. The presence of ECs in the finished drinking water of the Mathura and Agra DWTPs indicate the ineffectiveness of existing water treatment technologies in removing these compounds. During monsoon, the runoff from catchment area increases the concentration of the ECs. The study thus highlights the extensive and consistent use of pharmaceuticals and the need to upgrade the existing wastewater treatment facilities to remove the trace contaminants from domestic and industrial sewage. In India and developing countries with high population density, where partially treated water with high EC concentration is used for agriculture, possibly leading to contamination of the human food chain, the need of the hour is investment in advanced treatment techniques to effectively degrade a range of ECs in drinking water or waste water treatment plants to biologically non active forms or

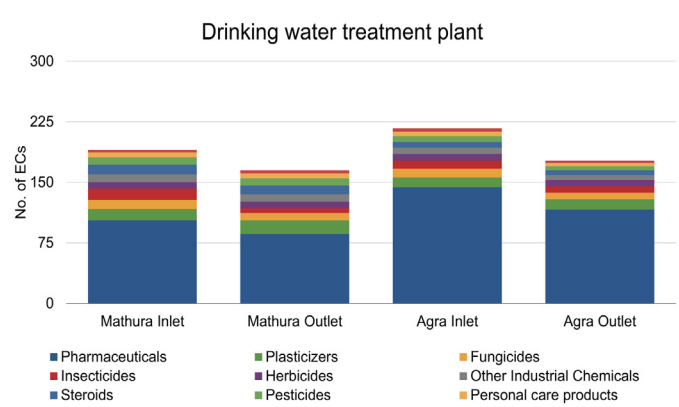


Fig. 4. Distribution of ECs in the inlet and outlet of Mathura and Agra DWTP.

levels. The current study indicates that rather than a targeted study of a handful of ECs, it is necessary to investigate for a wide range of ECs present and persistent in the environment, followed by administrative action to limit the use of such ECs.

CRediT authorship contribution statement

Pinakshi Biswas: Sampling, Method development, Analysis, Validation, Writing. Bhanu Prakash Vellanki: Conceptualization, Curation of data, Writing.

Declaration of competing interest

The authors declare that there are no known competing interests.

Acknowledgement

This research (FAME Project, Fate and Management of Emerging Contaminants) has been financed by two funding programs. One by Department of Science and Technology, Government of India and UK Natural Environment Research Council under India-UK Water Quality Program under grant number DST-1187-CED. Another by National Buildings Construction Corporation Limited India under Project no. NBC-933-CED. We acknowledge the contribution of Dr. Absar Ahmad Kazmi for securing the funding for the research. We also acknowledge the help of Dr. Ankur Rajpal and Dr. Muntjir Ali (Indian Institute of Technology, Roorkee) towards the present research.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2021.146741.

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